MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1907.3
STANDARD **Motion Imagery Metadata (MIMD): Payload**	**2025-06-11**

1 Scope

The Motion Imagery Metadata (MIMD) Model [1] includes information about Payloads hosted by Platforms (see MISB ST 1905 [2]). A Payload is a device or system of devices attached to a Platform for accomplishing the objectives of the system. Devices include general purpose intelligence gathering systems such as visible light cameras, infrared cameras, SAR, ESM, LIDAR, HSI, MSI, and acoustic devices, for example, a Motion Imagery system with its lenses, filters, and sensor arrays. The Payload model currently does not include weapon systems. The scope of this standard describes the modeling information about payloads with an initial focus on Geo-Intelligence Sensors. Other payload types, e.g., acoustical devices, are future extensions.

2 References

[1] MISB, *ST 1901.3 - Motion Imagery Metadata (MIMD): Modeling Rules,* 2025-06-11.
[2] MISB, *ST 1905.3 - Motion Imagery Metadata (MIMD): Platform,* 2025-06-11.
[3] MISB, *MISP-2025.1: Motion Imagery Handbook,* 2025-06-11.
[4] MISB, *ST 1906.3 - Motion Imagery Metadata (MIMD): Staging System,* 2025-06-11.
[5] MISB, *ST 1904.3 - Motion Imagery Metadata (MIMD): Base Attributes,* 2025-06-11.
[6] MISB, *ST 1908.3 - Motion Imagery Metadata (MIMD): Imager System,* 2025-06-11.
[7] MISB, *ST 1903.3 - Motion Imagery Metadata (MIMD): Model,* 2025-06-11.

3 Revision History

Version	Approved Date	Change Description
ST 1907.3	2025-06-11	• Added more background information for Correspondence Points; allowing sample array coordinates to extend beyond the sample array (i.e., off image coordinates) ○ Defined Image_Center type ○ Added new requirement -08 to place the image center correspondance (or copy of) in the GeoIntelligenceSensor's correspondances list. • Expanded the capabilities of Correspondence Groups ○ Defined sensor array shapes: points, polygon, polyline & ellipse ■ Added ellipsoid class ○ Defined rectangle, centroid & boundary as Correspondence Group properties ■ Added Geo-Rectangle property ■ Clarified definitions of rectangle and boundary as being minimal bounding areas on the image ■ Added new requirement -06 on boundary point order ○ Changed footprint definition to use polygon shape instead of boundary shape ■ Added ellipsoid-based footprint ■ Added new requirement -07 on footprint points • Added geoIntSensorParent attribute to Correspondence and CorrespondenceGroup to allow tracing back to the originating GeoIntelligenceSensor • Deprecated Payload class's name; name attribute is part of the Device Identity class, via the deviceId attribute • Deprecated GeoIntelligence class's name; name attribute is part of the Device Identity class, via the deviceId attribute • Added ontologyTermSuffix attribute to object class • Added gscpoStage attribute to GeoIntelligence class ○ Retired requirement -01 & replaced with new requirement -05 • Editorial changes to requirement -03

4 Definitions of Terms, Acronyms, and Initialisms

4.1 Terms

The following lists the terms and their definitions this document uses.

Correspondence	An interior or exterior relationship between the super-scene and the sample-array-plane
Correspondence Group	An imagery region-of-interest (see Section 6.2.2) containing at least one Correspondence Point
Correspondence Point	A single point providing either a Scene location (e.g., latitude, longitude, elevation), an image location (row, column image location), or both (which correspond to each other)
Exterior Correspondence	A correspondence point in exterior-scene

Exterior-Scene	The part of the super-scene which is not apart of the scene (super-scene minus scene); a subset of the super-scene and a mutually exclusive subset of scene
Interior Correspondence	A correspondence point in the scene which relates to a sample point in an image
Multipoint Correspondence Group	A Correspondence Group with two or more Correspondence Points
Sample-Array-Plane	A geometric plane co-planar to the sample array
Scene	The space in the physical world imaged by a sensor; a subset of the super-scene
Single Point Correspondence Group	A Correspondence Group with one Correspondence Point, and usually, a rectangle or radius
Super-Scene	The space in the physical world a sensor can or is imaging; the superset of the scene

4.2 Acronyms and Initialisms

The following lists the acronyms and initialisms this document uses.

GSCPO	Geo-Intelligence Sensor Common Pointing Orientation
HAE	Height Above Ellipsoid
IRI	Internationalized Resource Identifier
LRF	Laser Range Finder
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
ROC_Unknown	Report-on-Change Unknown state
ST	Standard
UML	Unified Modeling Language

5 Introduction

Several types of Payloads hosted on Platforms provide distinct functions to achieve system and mission objectives. One subset of Payload types are sensors that gather Geographical based Intelligence data on an image scene (see MISB Motion Imagery Handbook [3] for background on Sensors and Scene). A Geo-Intelligence Sensor is a device which measures active or passive Scene energy with a defined correspondence between the sensed data and the Scene.

While Payload devices share common attributes, each Payload device also defines specific metadata for their phenomenologies and methods of data collection. This standard includes the model for frame imagers (visible light and infrared), and Laser Range Finder (LRF); future revisions will include models for other Payload devices.

6 Overview

The Payload model is the combination of the physical Payload description and the resulting operational results in the form of Correspondences, Correspondence Groups and Tracks.

6.1 Payload Description

A Payload is a self-contained set of one or more devices along with its supporting hardware and software. The Payload is a physical component attached to (or riding on) a Platform (defined in MISB ST 1905) and may or may not need external power, communications, or controls. The MIMD Model provides a list of Payloads for a Platform, where each Payload may have one or more devices. Currently, the Payload only supports Geo-Intelligence devices. The Payload may define a reference point and orientation using Stages (defined in MISB ST 1906 [4]), which may provide the basis for the positioning and orientation information in each Payload device. The Payload position and orientation may be absolute or relative to the Platform position and orientation.

Figure 1 shows several images of potential Payloads. Image (a) is a Payload consisting of a single sensor ball with multiple sensors (e.g., Visible light, Infrared, Laser Range Finder). Image (b) is a combination of sensors and gun system. Currently, the Payload model only supports metadata for the sensor component of this system and not the guns. Image (c) and (d) is a ground-based tracking system where each sensor may be a separate Payload attached to a turreted Platform.



(a)
(Approved by USN for public release)



(b)
(Approved by USN for public release)



(c)
(Photo credit White Sands Missile Range)



(d)
(Photo credit White Sands Missile Range)

Figure 1: Example Payloads

6.1.1 Geo-Intelligence Sensor Orientation

A Geo-Intelligence Sensor is a conceptual sensor which may be directional or "point" to a Scene or point-target, for example an imaging sensor or an LRF. For this model there are two types of pointing sensors: Single Point sensors and Multi-Point sensors. Single point sensors only collect

one data value in the sensor's pointing direction (e.g., LRF). Multi-Point sensors collect more than one point at a time in the pointing direction; the result is either a one- or two-dimensional array of data (e.g., imaging sensor).

When specifying the Staging (see MISB ST 1906) orientation of any Single Point or Multi-Point Geo-Intelligence Sensor the MIMD Model defines a **Geo-Intelligence Sensor Common Pointing Orientation (GSCPO)** for the final or terminal stage. In all cases (Single Point or Multi-Point) the GSCPO has the positive x-axis pointing in the direction from the sensor to the Scene or point-target. The x-axis aligns with the boresight of the pointing sensor for the given sensor. With Multi-Point Sensors an array of samples is available (either one or two dimensional), so there is enough information to define a Scene and therefore the y and z axes orientations. The positive y-axis points to the right side of the Scene or in the direction of increasing column numbers of the resulting image. The positive z-axis points downward towards the bottom of the Scene or in the direction of increasing row numbers of the resulting image. The x-axis points to the middle of the resulting image.

6.1.1.1 Sample Array

The Geo-Intelligence Sensor models its output as an array of samples, which may be a single point, one dimensional, or two dimensional. The GeoIntelligenceSensor class uses attributes nCols and nRows to define the sizes of the dimensions. If both nCols and nRows are greater than one (1) the array is two-dimensional with the given nCols and nRow dimension sizes. If either nRows or nCols is one (1) and the other value is greater than one the array is a single dimension. If both nRows and nCols are one (1) then the array is a single point. Figure 2 illustrates the different sample array configurations.

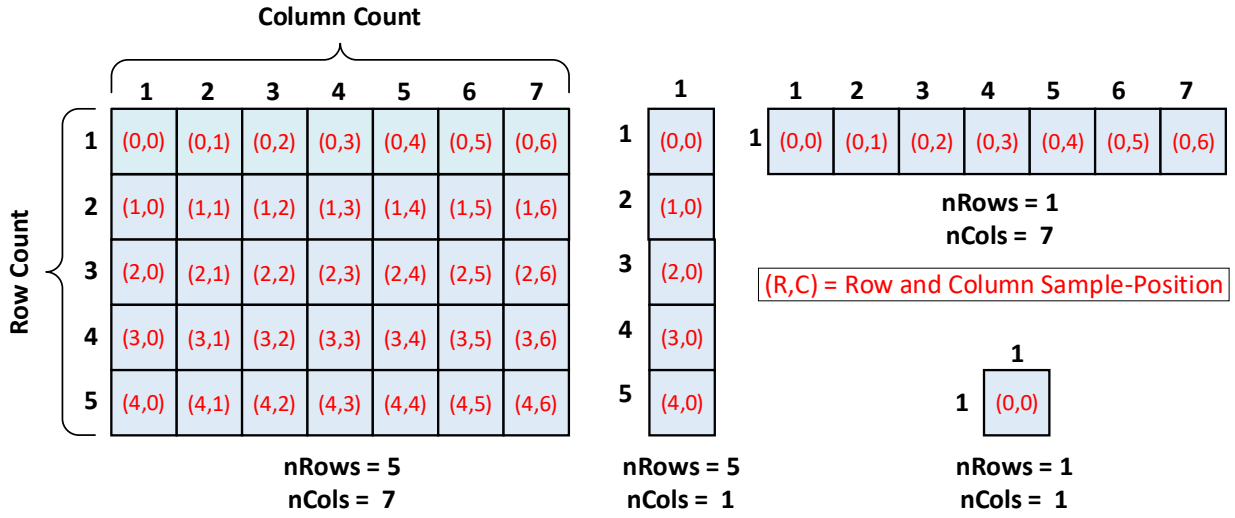


Figure 2: Sample Array Examples

Each sample in the array has a zero-based sample-position, which Figure 2 shows as red text.

The MIMD standards use sample-coordinates which are a refinement of the sample-position. Sample-coordinates specify both the sample-position within the array and a location within the sample itself. Sample-coordinates are real values and zero based, with the fractional value of the

coordinate being the location within the sample relative to the upper left of the sample. For example, the coordinate (0.0, 0.0) is the upper left corner (.0, .0) of the sample with sample-position (0, 0) (the upper left corner of the array). The sample-coordinate of the center of a sample has an offset of 0.5 in both the row and column direction, i.e., (*.5, *.5) where * is a given sample-position. Figure 3 illustrates a 5 row by 7 column sample array and highlights several sample-coordinates. Figure 3 shows the origin of the array at (0.0, 0.0) in the upper left of the sample array; the center of the first pixel in the array is at sample-coordinate (0.5, 0.5); the sample-coordinate for the center of the sample at row 2, column 3 is (2.5, 3.5). The sample-coordinate at (3.8, 5.7) illustrates an example computational result which did not resolve to the center of a sample. This illustration also shows the row and column axis, which represents the mathematical description of the sample array.

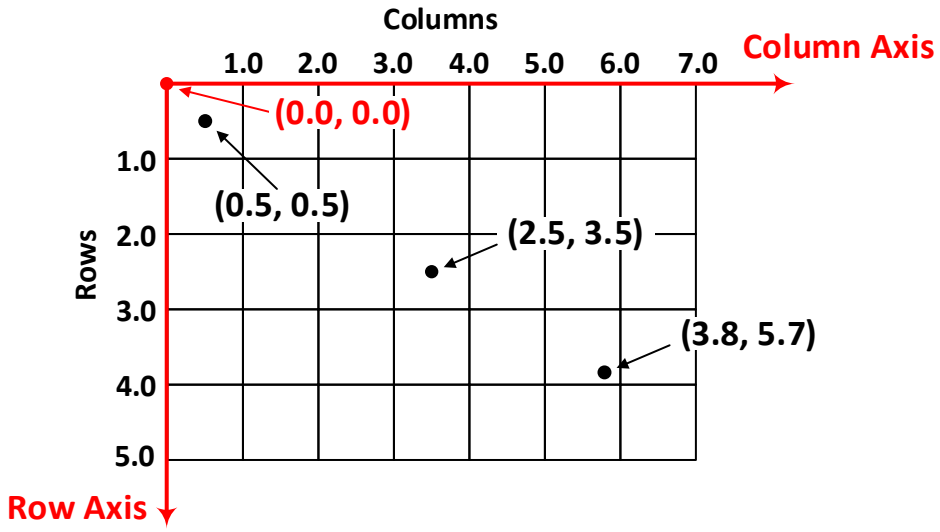


Figure 3: Example of a 5x7 Sample Array with Example Sample-Coordinates

The mathematical interval range of row coordinates is [0, nRows), where zero is a closed endpoint of the interval and nRows is an open interval (non-inclusive in the range); however, this standard defines the range as a fully closed interval, i.e., [0, nRows]. This means the maximum value of a row coordinate is nRows with no fractional value, e.g., if nRows is 5, the max value is 5.0, so 5.01 is an unallowed coordinate value. Similarly, the column interval range is [0, nCols].

6.2 Operational Results

In addition to the visual aspect of Motion Imagery, a major component of usability is the resulting operational metadata. This metadata describes items of interest in the Motion Imagery at three levels. The first level is single points of information which may be in the image, in the "Scene" or both; this first level of information is the Correspondence Point (Section 6.2.1). The second level is a conceptual area either in the image or in the scene, a conceptual area may or may not include a list of Correspondence Points; the second level of information is the Correspondence Group (Section 6.2.2). The third level is temporal tracking of Correspondence Groups in the image, the Scene or both; the third level of information is the Track (Section 6.2.3).

6.2.1 Correspondence Point

A correspondence is a relationship between the physical world and the sample array. Using a projective model, the sample array projects sample points from the perspective center onto the physical world; therefore, each image point projects onto a portion of the physical world. The space in the physical world the whole sample array projects onto is the sample array's "scene". The space in the physical world the sample array **can** project onto is the super-scene; the super scene is a "superset" of the scene. The part of the super-scene which is not a part of the scene, is the exterior-scene. Therefore, the super-scene = scene + exterior-scene. The "physical world" depends on the use-case and environment the sensor is employed in, e.g., the physical world can be the earth, the inside of a building, an object in the sky, etc.

Figure 4 illustrates a perspective model with the perspective center, sample array (image), and physical world on the earth (green). The scene is the brown area defined by the projection from the perspective center, through the image to the earth; the super-scene is the whole physical world, both brown and green; and the exterior-scene (green) is the super-scene area minus the scene area (brown), i.e., the sensor does not "see" exterior-scene points.

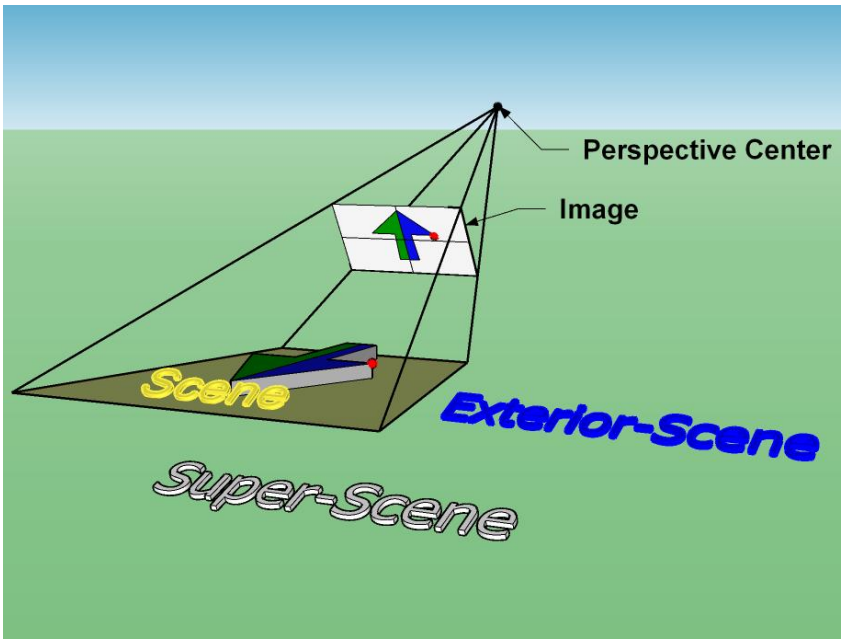


Figure 4: Scene, Exterior-Scene, and Super-Scene

For MIMD Correspondence Points, the sample array is the basis of a sample-array-plane, which is a geometric plane co-planar with the sample array, i.e., with the same origin and same 3D orientation as the sample array itself. This standard defines points within the sample array as interior points and points not in the sample array, but on the sample-array-plane, as exterior points.

A MIMD Correspondence Point is a relationship between the super-scene and the sample-array-plane. An **interior** correspondence is where a point in the scene corresponds to a sample array's sample point (e.g., red dot in Figure 4). An **exterior** correspondence is a point in the exterior-scene corresponding to a point in the sample-array-plane. A correspondence point may define the

sample-array-plane position (including negative sample array coordinates, or coordinates greater than the sample array size), super-scene position, or both. Depending on the relative position of the perspective center, perspective axis orientation, to the super-scene, not all exterior-scene points will project onto the sample-array-plane.

The correspondence relationship may be measurable, e.g., a geographical location (any 3-space on, above, or below the reference ellipsoid), or non-measurable, e.g., an object of interest clearly in the Scene but with an unknown geographical scene location. An example of a measurable correspondence is an LRF distance measurement between the LRF system and a point on the ground/surface[1] or object in the air, along the laser pointing direction. Figure 5 illustrates an LRF ranging a point on a Scene target (the arrow). The Range value is the distance between the LRFs position and a point in the direction the LRF is oriented, in this case the corner point on the arrow.

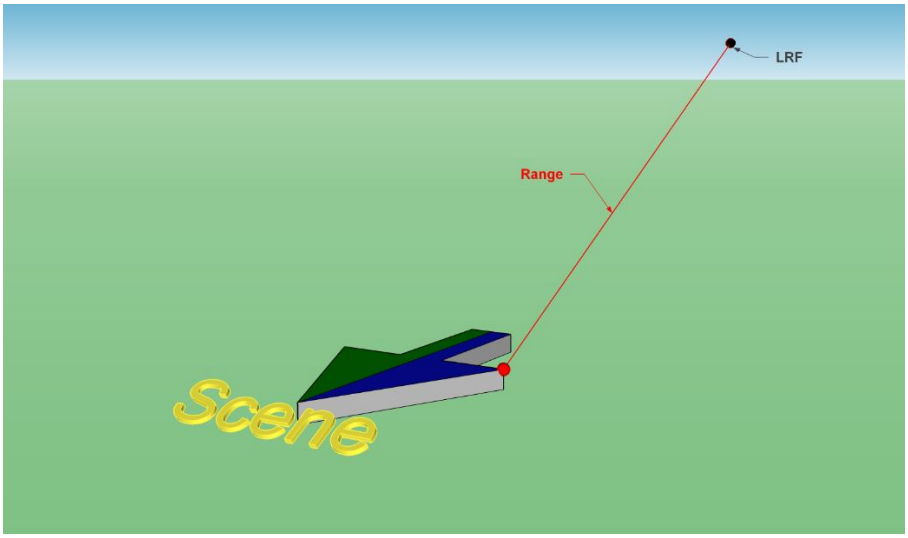


Figure 5: Example LRF Correspondence

Another example of a measurable correspondence is a sample point in a sample array corresponding to a geographical point in the Scene (e.g., a point on the ground, or a missile). Figure 6 illustrates a measurable Image Point to Target Point correspondence. The Image Point's sample location corresponds to the Target Point's position (e.g., latitude, longitude, HAE). The Target Point's position is typically algorithmically determined. The Geo-Intelligence sensor information provides the position and orientation of the Perspective Center and Image (i.e., sample array).

[1] Surface refers to the surface of a water body (i.e., ocean, river, lake)

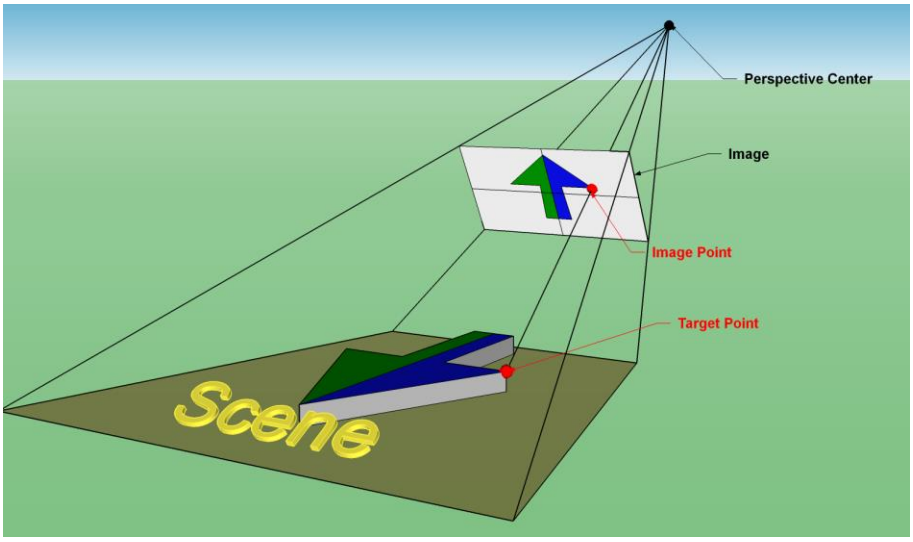


Figure 6: Example Measurable Image Correspondence

Figure 7 illustrates an example of a non-measurable correspondence. The sample location of the arrow are known but the Target Point position is unknown. Subsequent image processing may determine presence of an object with the correspondence between the pixel location and the arrow object.

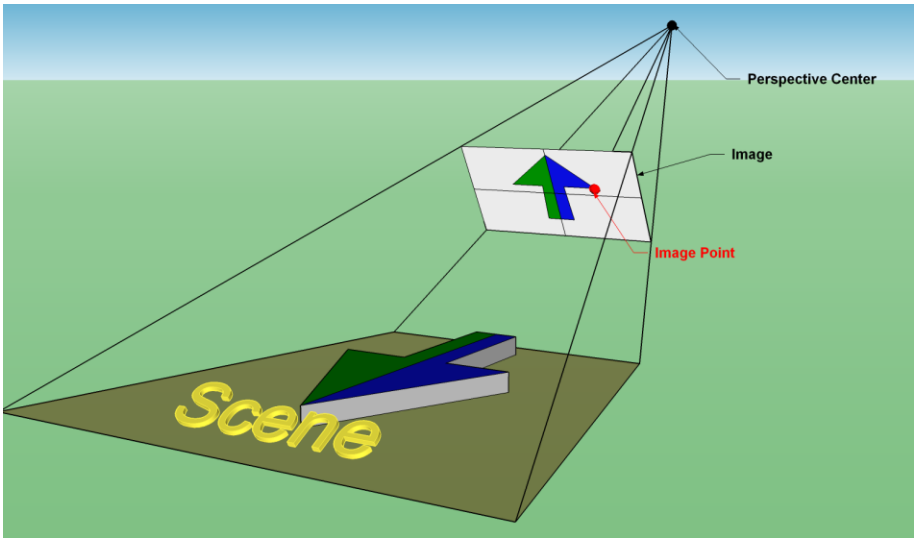


Figure 7: Example Non-Measurable Image Correspondence

There are two types of correspondences: Intra and Inter. Intra-correspondences are within a single sensor as illustrated in the previous figures above. Inter-correspondences are cross references between two or more sensors, such as an Imager and LRF, or two Imagers. Inter-correspondences may occur between sensors within a single Payload or between sensors in different payloads. Figure 8 illustrates a cross reference between a non-measurable image correspondence and a measured LRF correspondence. This illustration shows the Image's Perspective Center and the LRF are not co-located and therefore not bore sighted. The staging

data for the Perspective Center and LRF are very important modeling items to complete the geometry for target positioning.

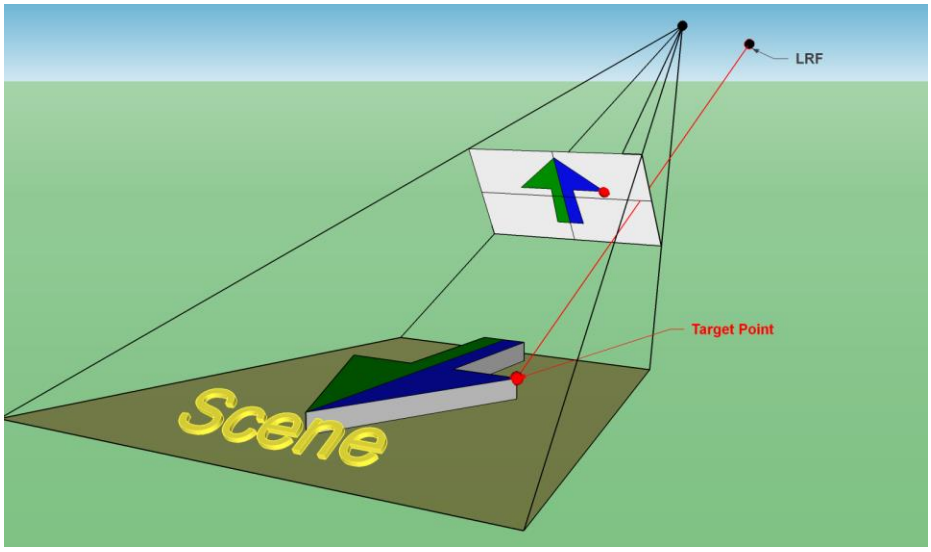


Figure 8: Example Inter-Correspondence between an Image and LRF

A useful set of default Image Intra-Correspondences are the image center point and four corners, or image footprint. Figure 9 illustrates these five default Image correspondences. The red line intersects the image center and corresponds to the scene point in the middle of the image. The four corners of the image correspond to four points on the ground/surface if the sensor is pointing downward. Additional points are the horizon line, if the Scene consists of ground/surface and sky, and the corners of an arbitrary bounding box.

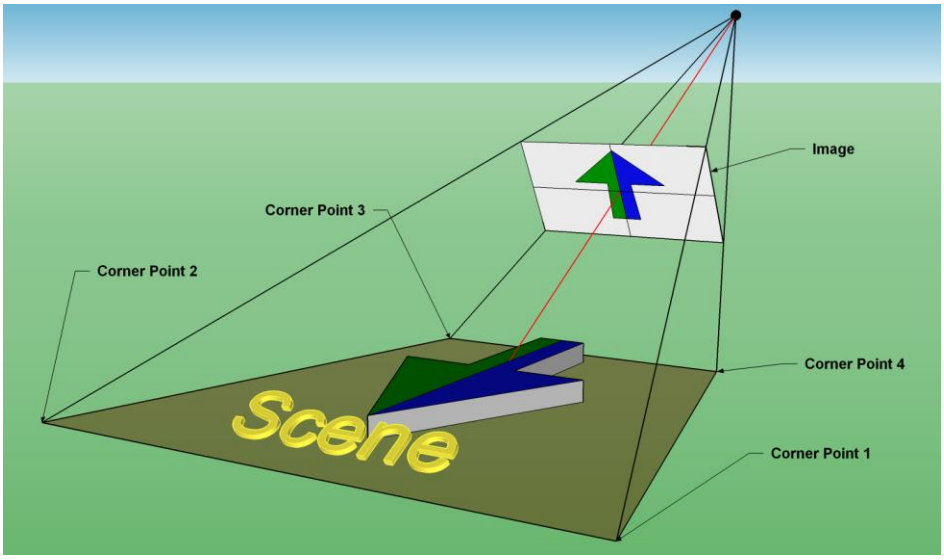


Figure 9: Default Image Correspondences

6.2.2 Correspondence Group

A Correspondence Group is a region-of-interest within the sensor-array-plane and/or super-scene where producers can: (1) collect related correspondence points, (2) define shapes using correspondence points, and (3) apply group properties. A Correspondence Group has a *type* attribute which declares the purpose of the group, see the CorrespondanceGroupType enumeration for details of each type.

1) The Correspondence Group contains an array of correspondence point references (*correspondences*) enabling producer software to collect separate points into one group and assign a purpose (i.e., *type*). For example, discrete points from one or more particle filters all relating to the same object or discrete points from different sensors about a target.
2) The Correspondence Group contains attributes to define the shape of an image object. The shapes are: *mask*, *point*, *ellipse*, *polygon*, and *polyline*. Section 6.2.2.1 provides details of the shapes.
3) With either (1) or (2), producers may define properties to aid MIMD readers processing and understanding. The properties are: *priority*, *confidence*, *color*, *intensity*, (bounding) *rectangle*, *centroid*, *boundary* (polygon), and *object*. Section 6.2.2.2 provides details of the properties

For example, a Correspondence Group may define the points for a moving object (e.g., car), or the shape of a background object (e.g., house), define a "tripwire" region (e.g., something which does not necessarily have a visual representation), or describe the four corners of the imagery footprint. A correspondence group may contain interior correspondence points, exterior correspondence points, or a mix of the two.

6.2.2.1 Correspondence Group Shapes

Correspondence group shapes define regions on the 2D sample-array-plane. The shapes are "defined" from the perspective of the sample-array-plane, but producers may project them to the super-scene and define Position instances.

A shape in the sample-array-plane may not preserve its appearance through a prospective transform to the super-scene, depending on the relative position of the perspective center, perspective axis orientation, to the super-scene. For example, a rectangle on a sample-array-plane may transform into a trapezoid when projecting into the super-scene. When performing the prospective transformation, some shape rules may not hold in the super-scene, e.g., a polygon projected (including with terrain elevation data) onto the super-scene may have line crossovers. The following summarizes the Correspondence Group sample-array-plane shapes.

- Mask – a list of positions in the sample array. Producers store MIMD masks as a bit array mask. E.g., list of all of the samples for a vehicle, or house in the imagery.
- Points – a list of one or more sample-array-plane points. The points have no relation to each other than being part of the Correspondance Group. For example, multiple points could represent the results of an object detection algorithm or a single point can be the basis for a point surrounded by a polygon (e.g., rectangle gate).

- Ellipse – an ellipse shape defined by a sample-array-plane center position, semi-major axis length (samples), semi-minor axis length (samples), and a rotation (radians). The Ellipse class supports circles (point and radius) by setting only the semi-major axis to the circle radius.
- Polygon – a closed simple polygon, i.e., the polygon lines do not cross, on the sample-array-plane. The polygon attributes list all its points in clockwise order and the final point does not need to match the first point, i.e., readers assume the final line of the polygon goes from the last point in the list to the first point in the list. The polygon shape provides the flexibility to define triangles, squares, rectangles, etc.
- Polyline – one or more connected line segments on the sample-array-plane.

6.2.2.2 Correspondence Group Properties

Producers may provide properties about the group to communicate to the reader administrative or summary information. The Correspondence Group class provides further details.

- Administrative information
  - Priority – importance of the group to the application, e.g., the primary vehicle for tracking.
  - Confidence – confidence the group has the correct points, e.g., the algorithm is 90% sure the points of an object are correct.
  - Color – the representative color of the group, e.g., tracking a red car.
  - Object – the result of an object identification process, e.g., object is a vehicle.
- Summary information
  - Rectangle – the minimal bounding box of all points and shapes in the sample-array-plane of the Correspondence Group (aligned with sample array rows and columns).
  - Geo-Rectangle – the minimal bounding box of all points in the super-scene of the Correspondence Group (aligned with latitude and longitude lines).
  - Centroid – the *centroid* attribute is the point within a group that represents the whole group, typically the “center of mass” of the all the sample-array-plane points.
  - Boundary – he *boundary* attribute is a list of sample-array-plane points which define the outer perimeter of the correspondence group.

6.2.3 Track

A Track is the management of an object over time. Correspondence points and correspondence groups only exist in the scope of the current Motion Imagery frame; they are valid for one instant of time. In subsequent frames the Correspondence points and groups may shift within the frame but producers do not maintain their history, frame to frame. In effect, a track provides a history of one or more Correspondence Groups over time.

7 Model Classes

Figure 10 shows the UML class diagram for the MIMD Payload. The Payload class includes attributes about the Payload and a list of Geographical Intelligence Sensors (GeoIntelligenceSensor), where each Sensor class includes a list of Correspondence classes.

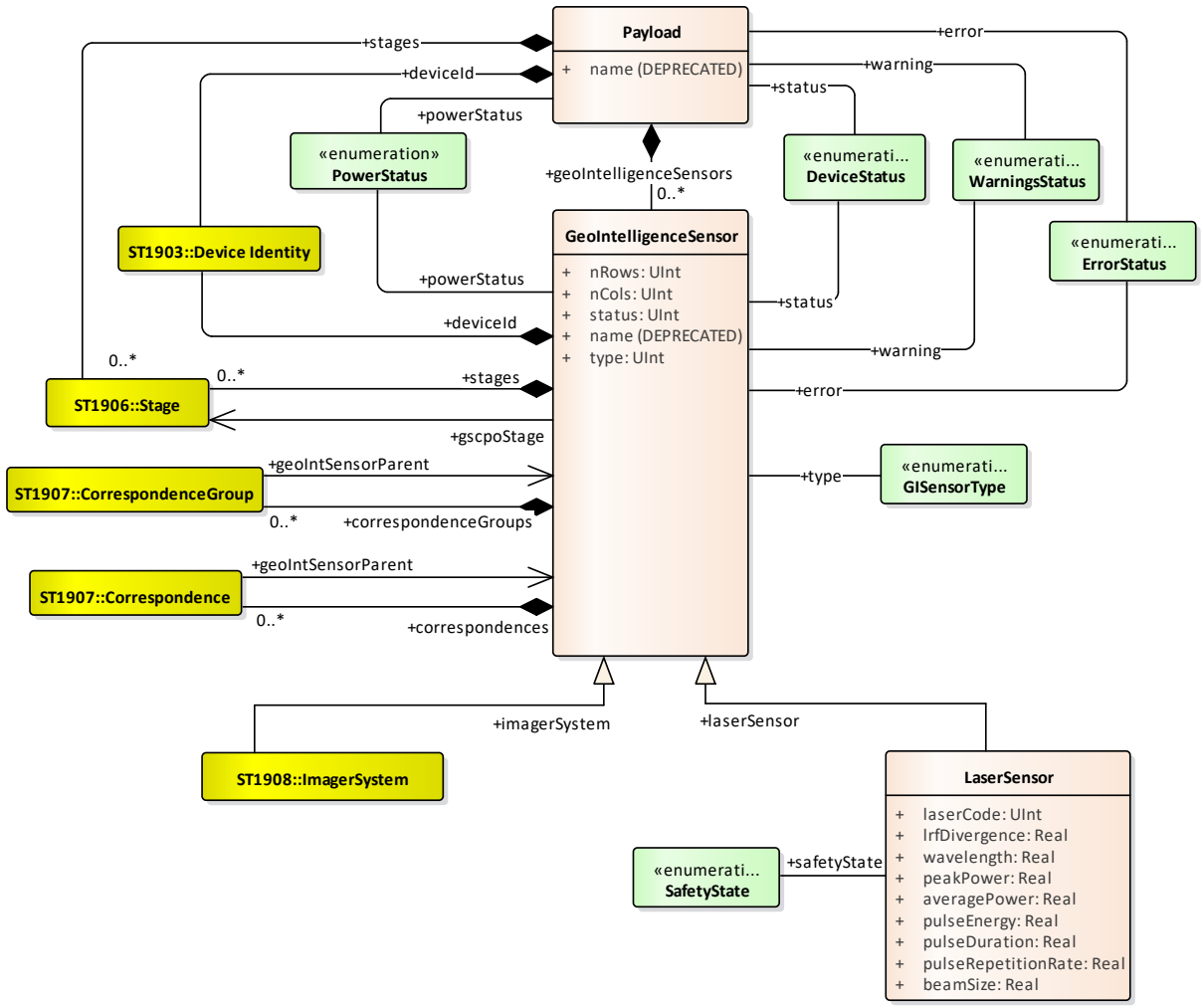


Figure 10: Payload Class Model

Per discussion in MISB ST 1901, MIMD classes support the Report-On-Change (reference the Motion Imagery Handbook) and include MIMD Base Attributes (MISB ST 1904 [5]), which use attribute identifiers in the range 1 through 32.

7.1 Payload Class

Table 1 lists the attributes for the Payload class.

Table 1: Payload Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	status	DeviceStatus	N/A	N/A	N/A	None	7.1.1
34	powerStatus	PowerStatus	N/A	N/A	N/A	None	7.1.2
~~35~~	~~name~~ **(DEPRECATED)**	~~String~~	~~N/A~~	~~N/A~~	~~100~~	~~None~~	~~7.1.3~~
36	stages	LIST<Stage>	--	--	N/A	None	7.1.4
37	geoIntelligenceSensors	LIST<GeoIntelligenceSensor>	--	--	N/A	None	7.1.5
38	warning	DeviceWarning	N/A	N/A	N/A	None	7.1.6
39	error	DeviceError	N/A	N/A	N/A	None	7.1.7
40	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.1.8

7.1.1 Attribute 33 – status

This attribute is an enumeration. Section 8.1 defines the DeviceStatus enumeration.

7.1.2 Attribute 34 – powerStatus

This attribute is an enumeration. Section 8.2 defines the PowerStatus enumeration.

7.1.3 ~~Attribute 35 – name~~ (DEPRECATED)

The *name* attribute of the DeviceIdenty class replaces the Payload class's *name* attribute; see the deviceId attribute.

7.1.4 Attribute 36 – stages

This attribute is a list of Stage classes. MISB ST 1906 defines the Stage class.

The Payload Stage is the position and orientation of a Payload reference point. Geo-Intelligence Sensors may use the Payload reference point to describe their stage's position and orientation. MISB ST 1906 defines the Stage class and all its child classes.

7.1.5 Attribute 37 – geoIntelligenceSensors

This attribute is a list of GeoIntelligenceSensor classes. Section 7.2 defines the GeoIntelligenceSensor class.

7.1.6 Attribute 38 – warning

This attribute is an enumeration. Section 8.3 defines the DeviceWarning enumeration.

7.1.7 Attribute 39 – error

This attribute is an enumeration. Section 8.4 defines the DeviceError enumeration.

7.1.8 Attribute 40 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

7.2 GeoIntelligenceSensor Class

Table 2 lists the attributes for the GeoIntelligenceSensor class.

Table 2: GeoIntelligenceSensor Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	nCols	UInt	--	--	N/A	Columns	7.2.1
34	nRows	UInt	--	--	N/A	Rows	7.2.2
~~35~~	~~name~~ **(DEPRECATED)**	~~String~~	~~N/A~~	~~N/A~~	~~100~~	~~None~~	~~7.2.3~~
36	type	GISensorType	N/A	N/A	N/A	None	7.2.4
37	status	DeviceStatus	N/A	N/A	N/A	None	7.2.5
38	stages	LIST<Stage>	--	--	N/A	None	7.2.6
39	correspondences	LIST<Correspondence>	--	--	N/A	None	7.2.7
40	imagerSystem	ImagerSystem	N/A	N/A	N/A	None	7.2.8
41	laserSensor	LaserSensor	N/A	N/A	N/A	None	7.2.9
42	correspondenceGroups	LIST<CorrespondenceGroup>	--	--	N/A	None	7.2.10
43	warning	DeviceWarning	N/A	N/A	N/A	None	7.2.11
44	error	DeviceError	N/A	N/A	N/A	None	7.2.12
45	deviceId	DeviceIdentity	N/A	N/A	N/A	None	7.2.13
46	gscpoStage	REF<Stage>	N/A	N/A	N/A	None	7.2.14

7.2.1 Attribute 33 – nCols

The *nCols* and *nRows* attributes specify the size of the sensor's output data in columns and rows. If both nCols and nRows are equal to one, the GeoIntelligenceSensor is a point source (e.g., LRF) sensor. If both are greater than one (1) the array is two-dimensional with the given nCols and nRow dimension sizes. If either attribute is one (1) and the other value is greater than one the array is a single dimension.

For systems which provide digital stabilization and/or digital zoom data in an ST 1908 [6] Imager Class instance, the nCols and nRows must match the dimensions of the stabilization / zoom window, see ST 1908's Imager class.

7.2.2 Attribute 34 – nRows

Refer to the description of attribute 33, *nCols*, for the details of this attribute.

7.2.3 ~~Attribute 35 – name~~ (DEPRECATED)

The *name* attribute of the DeviceIdenty class replaces the GeoIntelligenceSensor class's *name* attribute; see the GeoIntelligence's deviceId attribute.

7.2.4 Attribute 36 – type

This attribute is an enumeration. Section 8.5 defines the GISensorType enumeration.

7.2.5 Attribute 37 – status

This attribute is an enumeration. Section 8.1 defines the DeviceStatus enumeration.

7.2.6 Attribute 38 – stages

This attribute is a list of Stage classes. MISB ST 1906 defines the Stage class.

The *stages* attribute defines the stages which resolve to the pointing vector of the GeoIntelligenceSensor using the definition and orientation of the GSCPO in MISB ST 1906.

7.2.7 Attribute 39 – correspondences

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

7.2.8 Attribute 40 – imagerSystem

This attribute is a specialization relationship to the GeoIntelligenceSensor class. MISB ST 1908 defines the ImagerSystem class.

7.2.9 Attribute 41 – laserSensor

This attribute is a specialization relationship to the GeoIntelligenceSensor class. Section 7.3 defines the LaserSensor class.

7.2.10 Attribute 42 – correspondenceGroups

This attribute is a list of CorrespondenceGroup classes. Section 7.4 defines the CorrespondenceGroup class.

7.2.11 Attribute 43 – warning

This attribute is an enumeration. Section 8.3 defines the DeviceWarning enumeration.

7.2.12 Attribute 44 – error

This attribute is an enumeration. Section 8.4 defines the DeviceError enumeration.

7.2.13 Attribute 45 – deviceId

This attribute is a composite relationship to the Class DeviceIdentity. MISB ST 1903 defines the DeviceIdentity class.

7.2.14 Attribute 46 – gscpoStage

This attribute is a directed association (or reference) to Class Stage. MISB ST 1906 defines the Stage class.

The *gscpoStage* attribute is a reference to a single stage which defines the pointing vector of the GeoIntelligenceSensor. The stage uses the definition and orientation of the GSCPO in MISB ST 1906. If the *stages* attribute contains more than one terminal stage, producers must specify the gscpoStage attribute; otherwise, the default value for the gscpoStage attribute is the terminal stage in *stages* list.

Requirement	
ST 1907.3-05	Where producers include more than one terminal stage in the GeoIntelligenceSensor stages attribute, the producer shall set the gscpoStage to the terminal stage which defines the GSCPO for the GeoIntelligenceSensor.

7.3 LaserSensor Class

Table 3 lists the attributes for the Laser Sensor class. The LaserSensor Class supports all potential Laser sensors including Laser Range Finders (LRF), Laser illuminators, and LIDAR sensors (future). Small collections of range measurements may be in the LaserSensors's Correspondence list (e.g., LRF ranges). Larger collections of range measurements (e.g., LIDAR image) are usually in a separate data essence outside of the metadata, like sensor imagery.

Table 3: LaserSensor Class Attributes (extends GeoIntelligenceSensor)

Id	Name	Type	Min	Max	MLen	Units	Ref
33	laserCode	UInt	0	65 535	N/A	None	7.3.1
34	lrfDivergence	Real	0.0	--	N/A	mrad	7.3.2
35	wavelength	Real	0.0	--	N/A	µm	7.3.3
36	peakPower	Real	0.0	--	N/A	MW	7.3.4
37	averagePower	Real	0.0	--	N/A	KW	7.3.5
38	pulseEnergy	Real	0.0	--	N/A	J	7.3.6
39	pulseDuration	Real	0.0	--	N/A	s	7.3.7
40	pulseRepetitionRate	Real	0.0	--	N/A	Hz	7.3.8
41	beamSize	Real	0.0	--	N/A	mm	7.3.9
42	safetyState	SafetyState	N/A	N/A	N/A	None	7.3.10

7.3.1 Attribute 33 – laserCode

The *laserCode* attribute is the Pulse Repetition Frequency (PRF) code that laser designators generate to distinguish one laser beam from another. The Laser PRF code is a three- or four-digit

number only consisting of the digits one (1) through eight (8); the digits zero (0) and nine (9) are unallowed.

7.3.2 Attribute 34 – lrfDivergence

The *lrfDivergence* attribute is the Full Width at Half-Maximum (FWHM) angular measurement of the laser beam divergence.

7.3.3 Attribute 35 – wavelength

The *wavelength* attribute is the wavelength of the laser transmission.

7.3.4 Attribute 36 – peakPower

The *peak*P*ower* attribute is the instantaneous power at the laser sensor's output aperture. (e.g., 10 ns pulse at 10 mJ, peakPower = $\frac{10\ mJ}{10\ ns}$=1 MW)

7.3.5 Attribute 37 – averagePower

The *averagePower* attribute is the continuous power at the laser sensor's output aperture averaged over period of one second.

7.3.6 Attribute 38 – pulseEnergy

The *pulseEnergy* attribute is the power of a pulse.

7.3.7 Attribute 39 – pulseDuration

The *pulseDuration* attribute is length of time the laser is on during one pulse.

7.3.8 Attribute 40 – pulseRepetitionRate

The *pulseRepetitionRate* attribute defines the number of times the laser signal repeats per second.

7.3.9 Attribute 41 – beamSize

The *beamSize* attribute is the beam's diameter, at the exit aperture, where the power is at $1/e^2$ times the maximum power of the laser beam.

7.3.10 Attribute 42 – safetyState

This attribute is an enumeration. Section 8.6 defines the SafetyState enumeration.

7.4 CorrespondenceGroup Class

The Correspondence Group class is a collection of related correspondence points, see Section 6.2.2 for background. Figure 11 shows the UML for the Correspondence Group Class and its supporting classes.

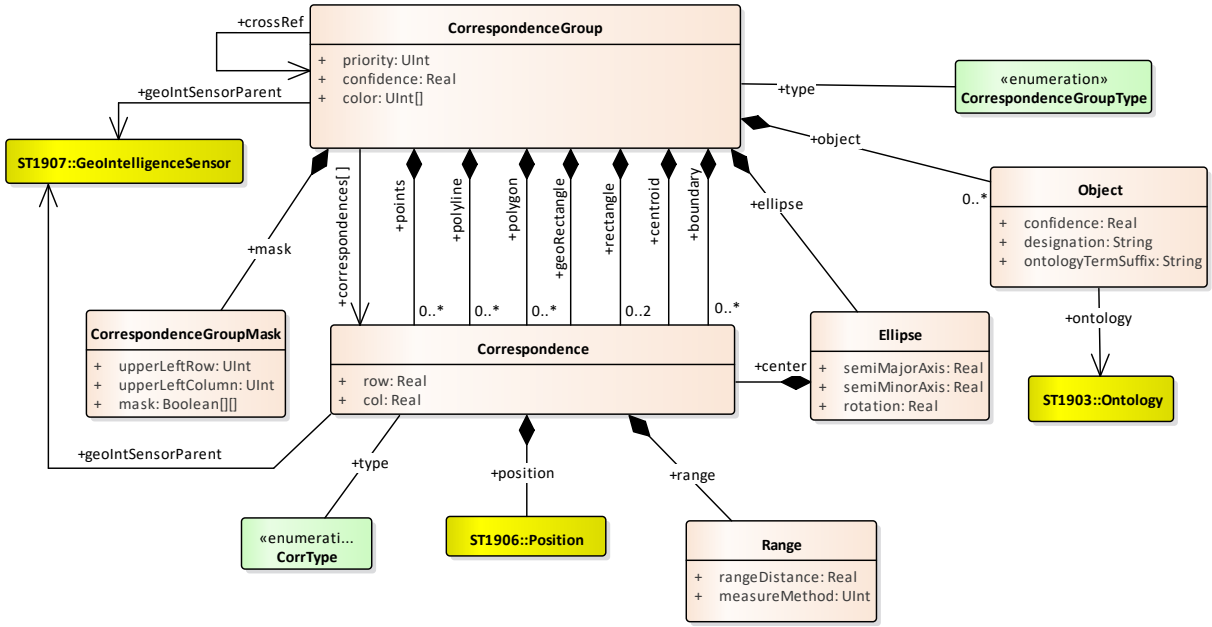


Figure 11: Correspondence Group Model

Per Section 6.2.2, a Multipoint Correspondence Group sets any one or a combination of the *correspondences* array attribute, *boundary* polygon attribute, or *mask* attribute. A Single Point Correspondence Group does not set any of the Multipoint Correspondence Group attributes but sets the centroid attribute. There are different requirements for the rectangle attribute when a Correspondence Group is a Multipoint Correspondence Group versus a Single Point Correspondence Group. Table 4 lists the attributes for a Correspondence Group class.

Table 4: CorrespondenceGroup Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	priority	UInt	--	--	N/A	None	7.4.1
34	confidence	Real	0.0	100.0	N/A	%	7.4.2
35	color	UInt[]	--	--	N/A	None	7.4.3
36	crossRef	REF<CorrespondenceGroup>	N/A	N/A	N/A	None	7.4.4
37	type	CorrespondenceGroupType	N/A	N/A	N/A	None	7.4.5
38	object	LIST<Object>	--	--	N/A	None	7.4.6
39	mask	CorrespondenceGroupMask	N/A	N/A	N/A	None	7.4.7
40	rectangle	LIST<Correspondence>	0	2	N/A	None	7.4.8
41	centroid	Correspondence	N/A	N/A	N/A	None	7.4.9
42	boundary	LIST<Correspondence>	--	--	N/A	None	7.4.10
43	algorithm	REF<Algorithm>	N/A	N/A	N/A	None	7.4.11
44	correspondences	REF<Correspondence>[]	N/A	N/A	N/A	None	7.4.12
45	points	LIST<Correspondence>	--	--	N/A	None	7.4.13
46	polygon	LIST<Correspondence>	--	--	N/A	None	7.4.14
47	polyline	LIST<Correspondence>	--	--	N/A	None	7.4.15
48	geoRectangle	LIST<Correspondence>	0	2	N/A	None	7.4.16
49	ellipse	Ellipse	N/A	N/A	N/A	none	7.4.17
50	geoIntSensorParent	REF<GeoIntelligenceSensor>	N/A	N/A	N/A	None	7.4.18

7.4.1 Attribute 33 – priority

The *priority* attribute defines the priority of the CorrespondenceGroup, where 1 is the highest priority. Zero (0) represents an unknown priority and is the default value if the priority attribute is ROC_Unknown.

7.4.2 Attribute 34 – confidence

The *confidence* attribute is the measure of CorrespondenceGroup confidence. For systems which use algorithmic methods to define a group of Correspondence Points as an "object" the confidence value provides a means of labeling how confident the points are the correct points, i.e., how confident the system is that the points are a part of the object. The *object* attribute (attribute 38) defines its own confidence measure for object classification. A value of 0.0 is essentially no confidence and a value of 100.0 is the highest confidence.

7.4.3 Attribute 35 – color

The *color* attribute defines the representative "color" of the group of points. The color attribute is an array of UInt values, where each value represents a color band. For monochrome images (e.g., Infrared), there is only one band and one value. For color imagery, the color attribute uses three bands, with the first element being the red value, second the green value, and the third the blue value (i.e., RGB). Systems using additional bands increase the length of the color array. The color band wavelengths match the second dimension of the wavelengths attribute of the

DetectorFilters class (see MISB ST 1908). The bitdepth attribute in the DetectorElement class (see MISB ST 1908) defines the range for the color measurements.

The method for determining the representative color is implementation dependent (e.g., average color of all points).

7.4.4 Attribute 36 – crossRef

This attribute is a directed association (or reference) to Class CorrespondenceGroup. Section 7.4 defines the CorrespondenceGroup class.

7.4.5 Attribute 37 – type

This attribute is an enumeration. Section 8.7 defines the CorrespondenceGroupType enumeration.

7.4.6 Attribute 38 – object

This attribute is a list of Object classes. Section 7.8 defines the Object class.

7.4.7 Attribute 39 – mask

This attribute is a composite relationship to the Class CorrespondenceGroupMask. Section 7.5 defines the CorrespondenceGroupMask class.

The *mask* attribute provides a means to specify exact pixel locations that form the Correspondence Group.

7.4.8 Attribute 40 – rectangle

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The *rectangle* attribute defines a minimal bounding rectangle of the sample-array-plane's points and shapes. The rectangle attribute defines the rectangle using two points, C_1 and C_2. The first corner, C_1, is the upper left corner of the rectangle and the second corner, C_2, is the lower right corner. The rectangle's sides are parallel with the sample array's edges. The minimal bounding rectangle is the smallest rectangle containing all the sample-array-plane points and shapes within the correspondence group.

Producers determine the minimum bounding box by computing r_{min}, c_{min}, r_{max}, and c_{max} for all the points and shapes within the sample-array-plane. The producer then combines the values to find the bounding two corners, i.e., $C_1 = (r_{min}, c_{min})$ and $C_2 = (r_{max}, c_{max})$.

Requirement(s)	
ST 1907.1-02	Where the CorrespondenceGroup's *rectangle* attribute list is set, the list shall contain two correspondences defining a rectangle.

ST 1907.1-03	Where the CorrespondenceGroup's *rectangle* attribute is set, the rectangle shall be a minimal bounding rectangle of all points and shape areas in the mask, points, ellipse, polygon, polyline, correspondences, boundary, and centroid attributes.

7.4.9 Attribute 41 – centroid

This attribute is a composite relationship to the Class Correspondence. Section 7.6 defines the Correspondence class.

The *centroid* attribute is the point within a group that represents the whole group. Typically, this value is at the center of the boundary; however, this is not a requirement.

7.4.10 Attribute 42 – boundary

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The *boundary* attribute is a list of Correspondences that are the outer perimeter of the sample points and shapes within the CorrespondenceGroup. The boundary is a closed simple polygon, (i.e., the polygon lines do not cross) which is a "hull" polygon (i.e., either a convex or concave hull) around all the sample points or shapes to encompass.

The polygon's point list is in clockwise order and the final point does not need to match the first point, i.e., readers assume the final line in the polygon goes from the last point in the list to the first point in the list.

Requirement	
ST 1907.1-04	Where the CorrespondenceGroup's *boundary* attribute list is set, the list shall define a closed simple polygon which is a hull polygon (i.e., either convex or concave hull) of all points and shape areas in the mask, points, ellipse, polygon, polyline, correspondences, boundary, and centroid attributes.
ST 1907.3-06	Where the CorrespondenceGroup's *boundary* attribute list is set, the producer shall list the points in clockwise order.

7.4.11 Attribute 43 – algorithm

This attribute is a directed association (or reference) to Class Algorithm. MISB ST 1903 defines the Algorithm class.

The *algorithm* attribute enables producers to state the algorithm that created the CorrespondenceGroup.

7.4.12 Attribute 44 – correspondences

This attribute is a directed association (or reference) to Class Correspondence. Section 7.6 defines the Correspondence class.

The *correspondences* attribute is an array of references to any of the GeoIntelligence classes correspondences list.

7.4.13 Attribute 45 – points

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The *points* attribute is a list of single points which producers define either in the sample-array-plane, super-scene, or both. Producers can use a point in conjunction with the polygon to provide a "gate" around point for tracking or other purposes.

See Section 6.2.2 for information on Correspondence Groups and their shapes.

7.4.14 Attribute 46 – polygon

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The polygon *attribute* defines a simple closed polygon within the sensor array. A simple polygon means there are no lines in the polygon crossing other lines. Each correspondence point in the list is a vertex of the polygon. Producers list the polygon's point list in clockwise order and the final point does not need to match the first point, i.e., readers assume the final line in the polygon goes from the last point in the list to the first point in the list. The polygon shape provides the flexibility to define triangles, squares, rectangles, etc.

Projecting the sample-array-plane points (2D) of the polygon to the scene points (3D) may not preserve the polygon shape or rules (i.e., lines may intersect).

See Section 6.2.2 for information on Correspondence Groups and their shapes.

7.4.15 Attribute 47 – polyline

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The *polyline* attribute defines a list of connected line segments in the sample-array-plane.

See Section 6.2.2 for information on Correspondence Groups and their shapes.

7.4.16 Attribute 48 – geoRectangle

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The *geoRectangle* attribute is equivalent to the rectangle attribute except the rectangle is aligned geodetically, i.e., the top and bottom of the rectangle is parallel to lines of latitude and the sides are parallel with lines of longitude. Producers define the geoRectangle using two corners, C_1 and C_2. C_1 is the upper left corner (most north and most west from the rectangle center) of the geo-rectangle and C_2 is the lower right corner (most south and most east from the rectangle center) of the geo-rectangle. The geo-rectangle is not a true rectangle as it represents an area on an elliptical surface. The geoRectangle is the minimum bounding box around all the scene points in the Correspondence Group.

For geospatial points, systems determine the minimum bounding box by computing lat_{min}, lon_{min}, lat_{max}, and lon_{max} from the Positions instances in each correspondence. When crossing the dateline make sure the min longitude and max longitude are set correctly. Bounding boxes do not include the North or South poles. Once the four min/maxes are found, producers

combine the values into the bounding two corners, i.e., $C_1 = (lat_{max}, lon_{min})$ and $C_2 = (lat_{min}, lon_{max})$.

Projecting the scene points (3D) into the sample-array-plane values may not preserve the rectangular shape or rules.

See Section 6.2.2 for information on Correspondence Groups and their shapes.

7.4.17 Attribute 49 – ellipse

This attribute is a composite relationship to the Class Ellipse. Section 7.10 defines the Ellipse class.

The *ellipse* attribute defines an ellipse on the sample-array-plane. A special case of an ellipse is a circle, so the Ellipse class attribute provides a point/radius shape within the Correspondence Group.

When computing the bounding rectangle or boundary polygon the ellipse edges are within the boundaries.

Projecting the ellipse to scene points (3D) may not preserve the shape.

See Section 6.2.2 for information on Correspondence Groups and their shapes.

7.4.18 Attribute 50 – geoIntSensorParent

This attribute is a directed association (or reference) to Class GeoIntelligenceSensor. Section 7.2 defines the GeoIntelligenceSensor class.

The geoIntSensorParent attribute enables a Correpondence Group to identify which sensor defines the group. This attribute is optional and for situations where multiple sensors are producing groups and there is a need to trace each group back to its originating sensor.

7.5 CorrespondenceGroupMask Class

The CorrespondenceGroupMask class defines a bit mask which identifies exact pixel locations to include in the Correspondence Group. The masking process first defines a set of pixels (e.g., an object) in an image, then computes a bounding box. The mask sets all the pixels-of-interest to true and all other pixels to false. Figure 12 illustrates the mask process with the pixels-of-interest being the vehicle (and its shadow). The masking algorithm computes the bounding box of the pixels-of-interest shown in red. The Image Chip is the same region enlarged for illustrative purposes. The Object Mask forms the mask array where its values are set to true for pixels-of-interest and false elsewhere. The Upper Left Corner row and column denote where, in the original image, the mask is located.

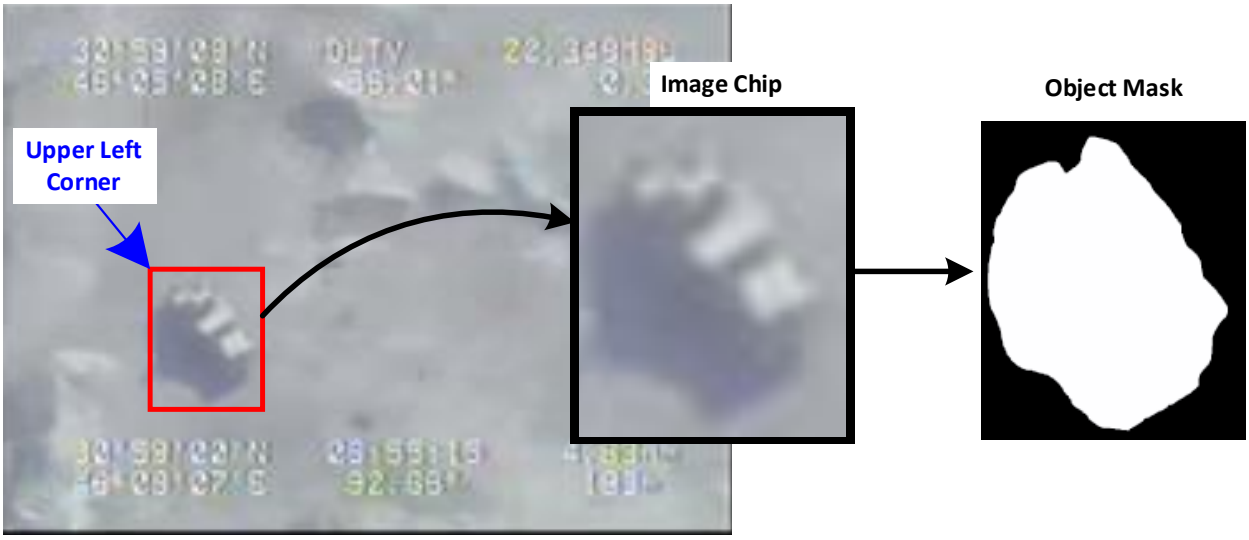


Figure 12:Illustration of Mask

Table 5: CorrespondenceGroupMask Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	upperLeftRow	UInt	--	--	N/A	None	7.5.1
34	upperLeftCol	UInt	--	--	N/A	None	7.5.2
35	mask	Boolean[][]	N/A	N/A	N/A	None	7.5.3

7.5.1 Attribute 33 – upperLeftRow

The *upperLeftRow* and *upperLeftCol* attributes state where in the image the mask is located. The values are in the range from zero to the nRows-1 and zero to nCols-1 for upperLeftRow and upperLeftCol respectively; nRows and nCols are attributes of the GeoIntelligenceSensor class.

7.5.2 Attribute 34 – upperLeftCol

Please refer to the description of attribute 33, *upperLeftRow*, for information about this attribute.

7.5.3 Attribute 35 – mask

The *mask* attribute is an array depicting the points-of-interest in a raster format. A value of true indicates a point-of-interest and a value of false indicates the point is not a point-of-interest.

7.6 Correspondence Class

Table 6 lists the attributes for the Correspondence class.

Table 6: Correspondence Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	col	Real	--	--	N/A	Columns	7.6.1
34	row	Real	--	--	N/A	Rows	7.6.2
35	type	CorrType	N/A	N/A	N/A	None	7.6.3
36	crossRef	REF<Correspondence>	N/A	N/A	N/A	None	7.6.4
37	position	Position	N/A	N/A	N/A	None	7.6.5
38	range	Range	N/A	N/A	N/A	None	7.6.6
39	geoIntSensorParent	REF<GeoIntelligenceSensor>	N/A	N/A	N/A	None	7.6.7

7.6.1 Attribute 33 – col

The *col* and *row* attributes together specify the correspondence sample-coordinates in the sensor data. See Section 6.1.1.1 for a description of sample-coordinates.

7.6.2 Attribute 34 – row

Refer to the description of attribute 33, *col*, for the details of this attribute.

7.6.3 Attribute 35 – type

This attribute is an enumeration. Section 8.9 defines the CorrType enumeration.

The *type* attribute provides meaning to the correspondence point. With the addition of CorrespondenceGroup class most of the CorrType enumeration values have been removed; however, the type attribute is maintained for future use.

7.6.4 Attribute 36 – crossRef

This attribute is a directed association (or reference) to Class Correspondence. Section 7.6 defines the Correspondence class.

The *crossRef* attribute is an Inter-Correspondence reference (to another correspondence), see Section 6.1.1 for overview of Inter-Correspondences.

7.6.5 Attribute 37 – position

This attribute is a composite relationship to the Class Position. MISB ST 1906 defines the Position class.

7.6.6 Attribute 38 – range

This attribute is a composite relationship to the Class Range. Section 7.7 defines the Range class.

7.6.7 Attribute 39 – geoIntSensorParent

This attribute is a directed association (or reference) to Class GeoIntelligenceSensor. Section 7.2 defines the GeoIntelligenceSensor class.

The geoIntSensorParent attribute enables a Correpondence to identify which sensor defines the correspondence. This attribute is optional and for situations where multiple sensors are producing correspondences and there is a need to trace each correspondence back to its originating sensor.

7.7 Range Class

Table 7 lists the attributes for the Range class.

Table 7: Range Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	rangeDistance	Real	0.0	--	N/A	m	7.7.1
34	measureMethod	MeasureMethod	N/A	N/A	N/A	None	7.7.2

7.7.1 Attribute 33 – rangeDistance

The *rangeDistance* attribute is the reported distance from the geo-intelligence sensor's terminal stage origin to the target point.

7.7.2 Attribute 34 – measureMethod

This attribute is an enumeration. Section 8.10 defines the MeasureMethod enumeration.

The *measureMethod* attribute defines the method for determining the rangeDistance attribute.

7.8 Object Class

The Object class records the results of an object classification process.

Table 8: Object Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	confidence	Real	0.0	100.0	N/A	%	7.8.1
34	ontology	REF<Ontology>	N/A	N/A	N/A	None	7.8.2
35	designation	String	N/A	N/A	100	None	7.8.3
36	ontologyTermSuffix	String	N/A	N/A	100	None	7.8.4

7.8.1 Attribute 33 – confidence

The *confidence* attribute is the measure of object classification confidence. The confidence value provides a means of labeling how confident the classification method is that the object is the identified object. A value of 0.0 is essentially no confidence and a value of 100.0 is the highest confidence.

7.8.2 Attribute 34 – ontology

This attribute is a directed association (or reference) to Class Ontology. MISB ST 1903 defines the Ontology class.

The *ontology* attribute references an ontology IRI prefix, which, when combined with the ontologyTermSuffix, provides the full IRI reference to the ontology term the producer assigns to this Object instance.

7.8.3 Attribute 35 – designation

The *designation* attribute is the human-readable term label which is the result of the object classification.

7.8.4 Attribute 36 – ontologyTermSuffix

The *ontologyTermSuffix* attribute is the latter half of a full ontology Internationalized Resource Identifier (IRI) reference to an ontological term. The ontology attribute provides a reference to the prefix and the ontologyTermSuffix provides the remaining part of the full IRI. For example, the Common Core Ontology (CCO) reference for the *Armored Personnel Carrier* term is *https://www.commoncoreontologies.org/ont00000467*. Each ontology term reference is the combination of two parts, an ontology prefix, and the term suffix. For example, the *https://www.commoncoreontologies.org/* is the ontology prefix (from the Ontology class) and the *ont00000467* is the term suffix. The prefix, by itself, is also an IRI; the suffix needs the prefix to be a meaningful IRI.

7.9 Track Class

Figure 13 shows the UML model for the Track Class. The Track class "tracks" Correspondence Groups over time providing a track history. Refer to the MIMD class in MISB ST 1903 [7] for the parentage of the Track class.

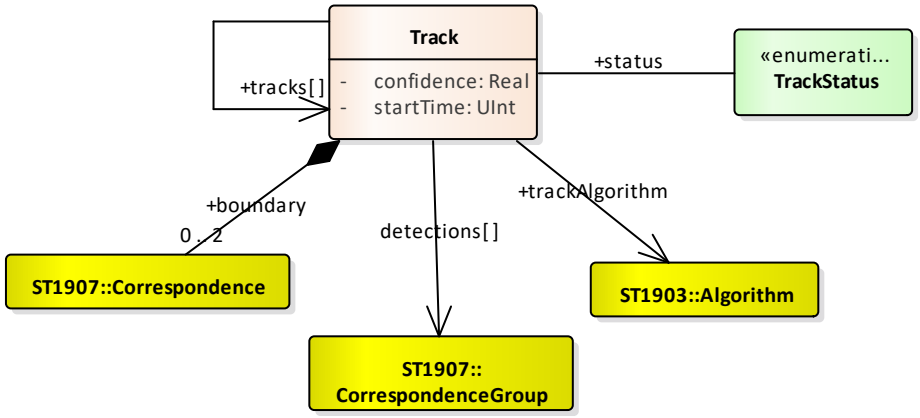


Figure 13: Track Class Model

Table 9 lists the Track class attributes.

Table 9: Track Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	startTime	UInt	--	--	N/A	ns	7.9.1
34	confidence	Real	0.0	100.0	N/A	%	7.9.2
35	trackAlgorithm	REF<Algorithm>	N/A	N/A	N/A	None	7.9.3
36	boundary	LIST<Correspondence>	0	2	N/A	None	7.9.4
37	detections	REF<CorrespondenceGroup>[]	N/A	N/A	N/A	None	7.9.5
38	tracks	REF<Track>[]	N/A	N/A	N/A	None	7.9.6
39	status	TrackStatus	N/A	N/A	N/A	None	7.9.7

7.9.1 Attribute 33 – startTime

The *startTime* attribute is the time when the track starts. When tracking software first assigns a Correspondence Group to a track this time value is a copy of the time of the first Correspondence Group in the track with the base class's timerOffset correction applied. For the life of the Track the startTime does not change, i.e., it is the Track's time of birth (birthtime).

7.9.2 Attribute 34 – confidence

The *confidence* attribute is the measure of track confidence. A value of 0.0 is essentially no confidence and a value of 100.0 is the highest confidence.

7.9.3 Attribute 35 – trackAlgorithm

This attribute is a directed association (or reference) to Class Algorithm. MISB ST 1903 defines the Algorithm class.

The *trackAlgorithm* attribute is a reference to an algorithm in the Algorithm list of the MIMD class. The track algorithm is the name of the algorithm producing the track along with other algorithm information (e.g., version, parameters).

7.9.4 Attribute 36 – boundary

This attribute is a list of Correspondence classes. Section 7.6 defines the Correspondence class.

The *boundary* attribute is a list of correspondence points which bound the track over time. If an object is moving, every update to the track could expand the boundary of the track.

7.9.5 Attribute 37 – detections

This attribute is a directed association (or reference) to Class CorrespondenceGroup. Section 7.4 defines the CorrespondenceGroup class.

The *detections* attribute is a list of detections over time.

7.9.6 Attribute 38 – tracks

This attribute is a directed association (or reference) to Class Track. Section 7.9 defines the Track class.

The *tracks* attribute is an array of Track references that relate to the current Track instance. For example, a tracking system tracks a car for some time then loses the track, sometime later a new track is started for an object and the system determines the object is the same car. The original car track (or tracks) may be associated with the current track via the tracks attribute.

7.9.7 Attribute 39 – status

This attribute is an enumeration. Section 8.8 defines the TrackStatus enumeration.

7.10 Ellipse Class

The Ellipse class defines an ellipse on the sample-array-plane using the center point (P), semi-major axis length (a), semi-minor axis length (b), and a rotation (θ).

The equation for an ellipse rotated by angle θ is:

$$\frac{((x - P_x)\cos(\theta) + (y - P_y)\sin(\theta))^2}{a^2} + \frac{((x - P_x)\sin(\theta) - (y - P_y)\cos(\theta))^2}{b^2} = 1$$

By setting P, a=b, and $\theta = 0$, the ellipse is a circle. The default value for the semiMinorAxisLength is equal to the semiMajorAxisLength and the default value for rotation is zero.

Table 10: Ellipse Class Attributes

Id	Name	Type	Min	Max	MLen	Units	Ref
33	center	Correspondence	N/A	N/A	N/A	None	7.10.1
34	semiMajorAxisLength	Real	--	--	N/A	Pixels	7.10.2
35	semiMinorAxisLength	Real	--	--	N/A	Pixels	7.10.3
36	rotation	Real	-PI	PI	N/A	rad	7.10.4

7.10.1 Attribute 33 – center

This attribute is a composite relationship to the Class Correspondence. Section 7.6 defines the Correspondence class.

The *center* attribute is the Correspondence point defining the sample-array-plane center location of the ellipse.

7.10.2 Attribute 34 – semiMajorAxisLength

The *semiMajorAxisLength* attribute is the length from the ellipse center to the ellipse along the major axis.

7.10.3 Attribute 35 – semiMinorAxisLength

The *semiMinorAxisLength* attribute is the length from the ellipse center to the ellipse along the minor axis.

7.10.4 Attribute 36 – rotation

The *rotation* attribute is the angle of counter-clockwise rotation of the semi-major axis from the x-axis (image sample row), e.g., zero is no rotation, $\frac{\pi}{2}$ is a 90 degree rotation up.

8 Model Enumerations

8.1 DeviceStatus Enumeration

Table 11: Enumeration Values for DeviceStatus

Id	Name	Description
0	Nominal	The device is in a nominal state
1	Stowed	The device is oriented in a manner that protects it or is in another temporary non-functioning state

8.2 PowerStatus Enumeration

Table 12: Enumeration Values for PowerStatus

Id	Name	Description
0	Off	The device is powered off
1	On	The device is powered on

8.3 DeviceWarning Enumeration

Table 13: Enumeration Values for DeviceWarning

Id	Name	Description
0	Nominal	No Warnings
1	Unspecified_Warning	The device is reporting a warning with unknown meaning
2	Over_Temperature	The device is over its maximum temperature
3	Under_Temperature	The device is under its minimum specified temperature

8.4 DeviceError Enumeration

Table 14: Enumeration Values for DeviceError

Id	Name	Description
0	Nominal	No Errors
1	Unspecified_Error	The device is reporting an error with unspecified meaning
2	Over_Temperature	The device is over its maximum temperature
3	Under_Temperature	The device is under its minimum specified temperature
4	Stowage_Error	The device has a problem while trying to stow or unstow

8.5 GISensorType Enumeration

Table 15: Enumeration Values for GISensorType

Id	Name	Description
0	EO	Electro Optical Sensor
1	LIDAR	LIDAR or LRF Sensor
2	RADAR	RADAR Sensor
3	SIGINT	Signal Intelligence Sensor
4	SAR	Synthetic Aperture Radar Sensor

8.6 SafetyState Enumeration

Table 16: Enumeration Values for SafetyState

Id	Name	Description
0	Safe	Laser is set to not be able to fire
1	Armed	Laser is ready to fire
2	Firing	Laser is actively propagating energy
3	Inhibited	Laser is prevented from firing for safety or other reasons (e.g., over temp)

8.7 CorrespondenceGroupType Enumeration

Table 17: Enumeration Values for CorrespondenceGroupType

Id	Name	Description
1	Background	Background Object Group Type
2	MTI	Moving Target Indicator Group Type
3	Footprint	The CorrespondenceGroup defines the image footprint and image center point, may include the horizon line as needed

8.7.1 Footprint

A footprint is the area a sample array projects onto the earth's surface. Producers store the footprint corners in the *polygon* or *ellipsoid* attribute, and set the *type* attribute to Footprint. The footprint's correspondences usually include the scene points (e.g., Position class metadata), and optionally includes sample array coordinates (e.g., row/col.

There are three types of footprints, full-footprint, horizon-footprint, and ellipsoid-footprint. A full-footprint is a footprint where the four corners of the image have geo-positions on the earth, so every image point represents a position on the earth. A partial-footprint is when the image has some area which does not represent points on the earth, i.e., the image includes the earth's horizon. Typically, the horizon is a line passing through the image from image edge to image-edge; however, producers can build more complex boundary polygon representations if necessary. An ellipsoid-footprint is where the producer desires to indicate the curvature of the earth, using an ellipse, for the footprint. Table 18 provides details of the footprint types.

Table 18: Footprint Types

Footprint type (shape attribute)	Number of Points	Description
Full-Footprint (polygon)	4	Moving clockwise around the image, the first footprint Correspondence point is the images upper left corner, the second footprint point is the upper right corner, the third point is the lower right corner, and the fourth point is the lower left corner. Because the image corners equal the footprint corners, producers do not need to specify the corners' image points; readers can derive them from the nCols and nRows attributes, i.e., corner one is at image coordinate 0,0; corner two is at row 0, image's nCols-1; etc.
Partial-Footprint (polygon)	3, 4, 5, or more	In the case where the upper left image corner does not project onto the earth (i.e., the image corner is above the horizon), the footprint corner point closest to the upper left image corner (via image coordinates) becomes the first point in the boundary definition. When the footprint has at least one point which does not represent the corner of the image, the producer must report the sample coordinates for all Correspondences in the boundary.
Ellipsoid-Footprint (ellipsoid)	N/A	Producers determine the center and orientation of the ellipsoid in the sample-array-plane, such that the curvature of the earth in the imagery matches the ellipse equation; it is likely the center will be an exterior correspondence point. Readers intersect the ellipse and the sample array bounds to determine the true footprint.

When a reader processes a footprint polygon, if the polygon points do not include sample coordinates, the reader can deduce the polygon is a Full-Footprint.

Figure 14 shows two footprint examples, the example on the left shows the footprint corners are equal to the image corners, with the corners labeled as the positions in the polygon. The example on the right shows an image with a horizon line (in red), in this case the footprint is five points; since point one is closer to the upper left corner than point two, point one becomes the starting point of the boundary list.

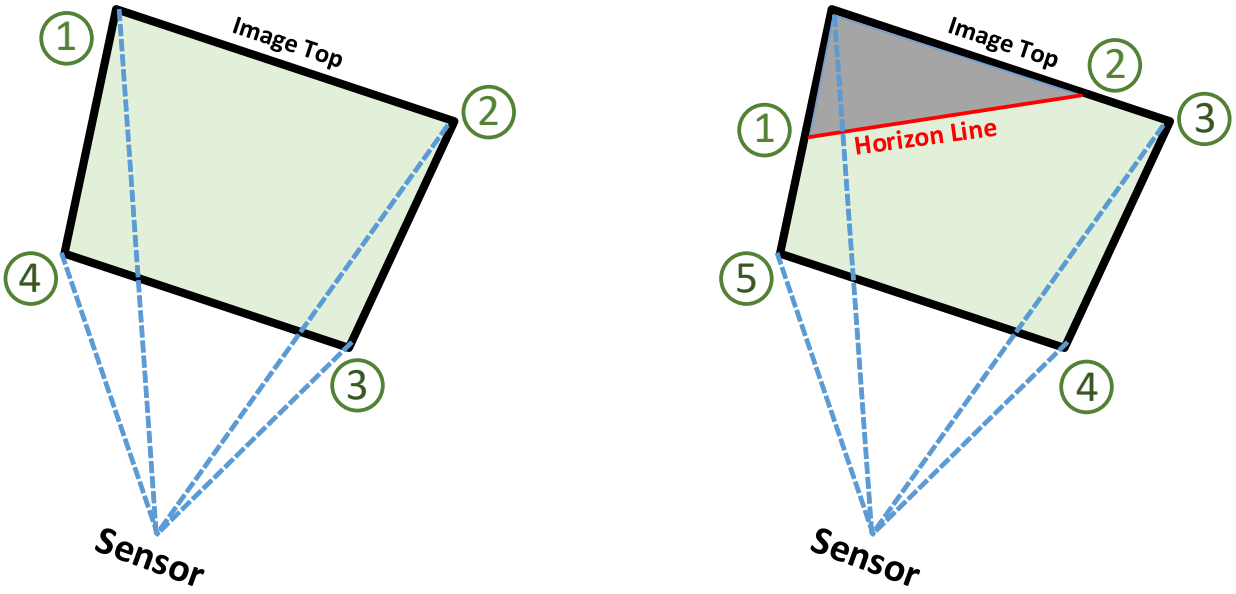


Figure 14: Footprint Examples

Requirement	
ST 1907.3-07	When a CorrespondenceGroup instance's type attribute is set to Footprint, where at least one of the polygon's points does not represent the sample array corner, the producer shall include the sample array coordinates within each polygon's Correspondence.

8.8 TrackStatus Enumeration

Table 19: Enumeration Values for TrackStatus

Id	Name	Description
1	Active	Target Track is active
2	InActive	Target track is not active
3	Dropped	Target no longer being tracked
4	Stopped	Target is not moving
5	Coast	Target lost but tracking on an estimated trajectory

8.9 CorrType Enumeration

The CorrType Enumeration provides additional information about the correspondence. With the addition of CorrespondenceGroup class most of the CorrType Enumeration values have been removed; however, the Enumeration is maintained for future use.

Table 20: Enumeration Values for CorrType

Id	Name	Description
0	General_Correspondence	Default value if not including type attribute in the Correspondence instance. If the type attribute is in the ROC_Unknown state, General Correspondence is assumed.
1	Image_Center	Indicates the correspondance position and/or image point (i.e., row,column) define the center of the image.

8.9.1 Image_Center

A correspondance with the *type* attribute set to Image_Center indicates the correspondance's position and/or image point defines the center of the image. For the case where only the position attribute is set, the reader assumes the image coordinates are at the middle of the image, i.e., $(col, row) = (\frac{nCol}{2}, \frac{nRow}{2})$. When the producer does provide the image coordinates, the coordinates may not be at the computed image center due to other factors the producer may adjust for (e.g., distortions, etc.). Figure 15 illustrates the image to scene relationship for the image center point. The figure shows the Image Center as a red ball on the image (in image coordinates) and the Position (yellow) which is a geo-coordinate.

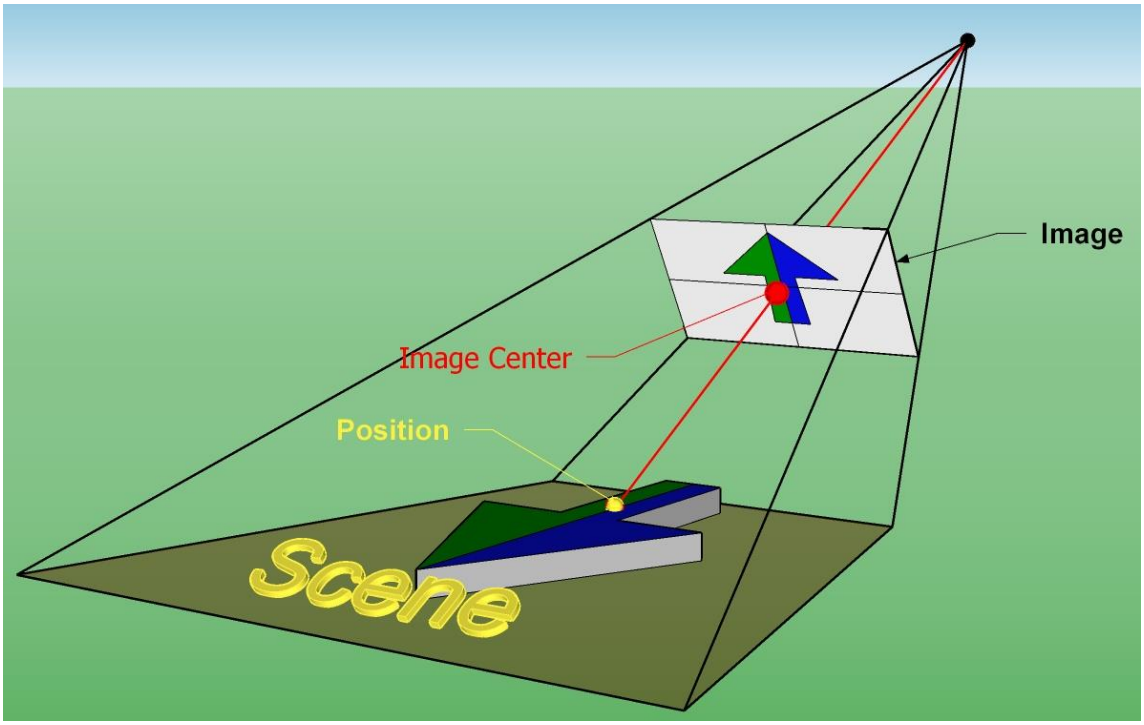


Figure 15: Illustration of Image_Center Correpondance Point

To aid readers in easily finding the image center correspondance within the model, when a producer creates an Image Center correspondance the producer inserts it, or a copy of it, in the GeoIntelligenceSensor's correspondence attribute list. This means the model can have multiple Image Center correspondences, e.g., a CorrespondenceGroup may contain an Image Center correspondence along with the GeoIntelligenceSensor's correspondence list.

Requirement	
ST 1907.3-08	When a Correspondence instance's type attribute is set to Image_Center, the producer shall include the Correspondence instance or a copy of the Correspondence instance in the GeoIntelligenceSensor instance's correspondence attribute list.

8.10 MeasureMethod Enumeration

Table 21: Enumeration Values for MeasureMethod

Id	Name	Description
0	Unknown	The method is not known. This is the default.
1	Measured	The range distance attribute's value is a measurement using a geo-intelligence sensor such as an LRF
2	Computed	The range distance attribute's value is the result of a computation

Appendix A Retired Requirements

Requirement	
ST 1907-01	Where the GeoIntelligenceSensor Stage attribute list is set, the terminal stage shall match the definition and orientation of the GSCPO. *(Staging system no longer requires single terminal stage per list; replaced with requirement 05)*